



**Ahmedabad
University**

CSE 623 Machine Learning

Weekly Report - 8

Group- Vector Minds

Team Members:

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Project Title

Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

Project Definition

Agriculture plays an important role in our lives, especially in a well-known agricultural country like India. It provides food security to a larger population; however, its productivity depends heavily on factors such as rainfall and seasonal weather conditions. Due to increasing climate variability, predicting crop yields often becomes difficult.

The aim of this project is to develop a predictive system that estimates crop yield based on provided agricultural and historical data. The project applies a classical machine learning regression model to analyze the effect of environmental factors on agricultural productivity. The project treats crop yield prediction as a supervised regression problem.

Where:

- Input variables: State, Crop type, Season, Crop Year, Area, Annual Rainfall, Fertilizer, Pesticide, and four engineered features — Fertilizer per Area, Pesticide per Area, Rainfall per Area, and Input Intensity
- Output variable (Target): Crop yield, which is production per unit area.

Week 8 Objectives

- Select the better model between Random Forest and Gradient Boosting by comparing their GridSearchCV optimization results in Week 6 and prepare it for deployment.
- Optimize both Random Forest and Gradient Boosting models through Bayesian Optimization using Optuna and compare its global optimal results with those obtained from GridSearchCV.
- Perform feature importance analysis to identify the most influential features involved in crop yield prediction for the final model.
- Check the error distribution of the final model to verify if it is symmetrical.
- Deploy the final model using joblib without training it again.
- Create a prediction function that predicts crop yield from an input sample.

Work Completed

Tasks performed during this final week involve choosing the best-performing model using the results of the past-week's tuning, performing further optimization using Bayesian Optimization, creating the explanation layer using the concept of feature importance, assessing prediction

errors, storing the model to be deployed and developing a fully accurate working prediction system.

Model Selection for Final Implementation

To select the final model, a programmatic approach was used to check the `best_score_` values obtained after applying GridSearchCV to optimize the Random Forest and Gradient Boosting models in Week 6. Because the Random Forest model had a higher cross-validated R^2 value, it was selected as the final model. First, the model name was displayed to confirm correct model selection, followed by other actions.

Bayesian Optimization using Optuna

In addition to the previous grid search using GridSearchCV that we did last week, we performed Bayesian Optimization this week using Optuna. Unlike GridSearchCV, Bayesian Optimization with Optuna does not evaluate all fixed hyperparameter combinations beforehand. Rather, it implements the Tree-structured Parzen Estimator (TPE) sampling strategy, a probability-based model that learns from previous attempts and samples only the most promising regions of the hyperparameter space. This method has proven to be highly efficient compared to grid search but also more accurate than random search. For the Random Forest model, Optuna evaluated 40 trials using the parameters `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, and `max_features`. As for the Gradient Boosting model, Optuna conducted 30 trials using the parameters `n_estimators`, `learning_rate`, `max_depth`, `subsample`, and `min_samples_split`. We then plotted a Bayesian convergence chart, which showed improvements in R^2 performance across iterations.

Feature Importances

The importance values of the final model's features were extracted from its `feature_importances_` attribute. The names of each feature after applying the OneHotEncoder transformation were collected to properly identify both categorical and numerical features. The bar plot depicting the importance values included all available features, as well as the four new ones introduced during Week 5: Fertilizer per Area, Pesticide per Area, Rainfall per Area, and Input Intensity. This plot shows that, as in previous weeks, Area, Annual Rainfall, Fertilizer, and Crop remain the key determinants of crop yield.

Distribution of Prediction Errors

First, the difference between the actual and predicted values was calculated as an error. Then, this distribution of errors was depicted as a histogram with a KDE overlay, with the zero point marked by a reference line, in the folder of graphs. Symmetry around the point of zero was checked to evaluate whether the distribution is bell-shaped around this point, indicating that the model produces unbiased predictions.

Actual vs Predicted Scatter Plot and Learning Curve

A plot showing the actual vs predicted values of our models in the final models has been created to validate the prediction ability of these models against the perfect-fit line which is along the diagonal. Also, the learning curve for the models used in the final analysis is presented, with the training and validation R^2 scores plotted at different training set sizes.

Final Model Comparison

A detailed comparison table and bar graphs have been made to compare all the models considered in Week 7. The models compared include Random Forest and Gradient Boosting with Grid Search Cross Validation and Bayesian Optimization from Optuna. The metrics considered are mean absolute error (MAE), root mean squared error (RMSE), and R^2 .

Conclusion

Last week marked the successful completion of building a model that effectively predicts crop yields. The final model was chosen objectively based on its performance after cross-validation in week six, and then compared with other Bayesian-optimized models. Feature importance has been used to analyze the explainability of the final model, while its reliability has been assessed through error distributions and comparisons of actual versus predicted values. Finally, the model has been deployed for real-world use, with a prediction function developed for easy integration. Overall, throughout all seven weeks, the entire machine learning process cycle has been covered, starting with data cleaning, exploratory data analysis, feature engineering, model training, hyperparameter tuning using GridSearchCV and Bayesian Optimization, ensembling and cross-validation to finally selecting the best model and creating an explainable, deployable and predictive system.